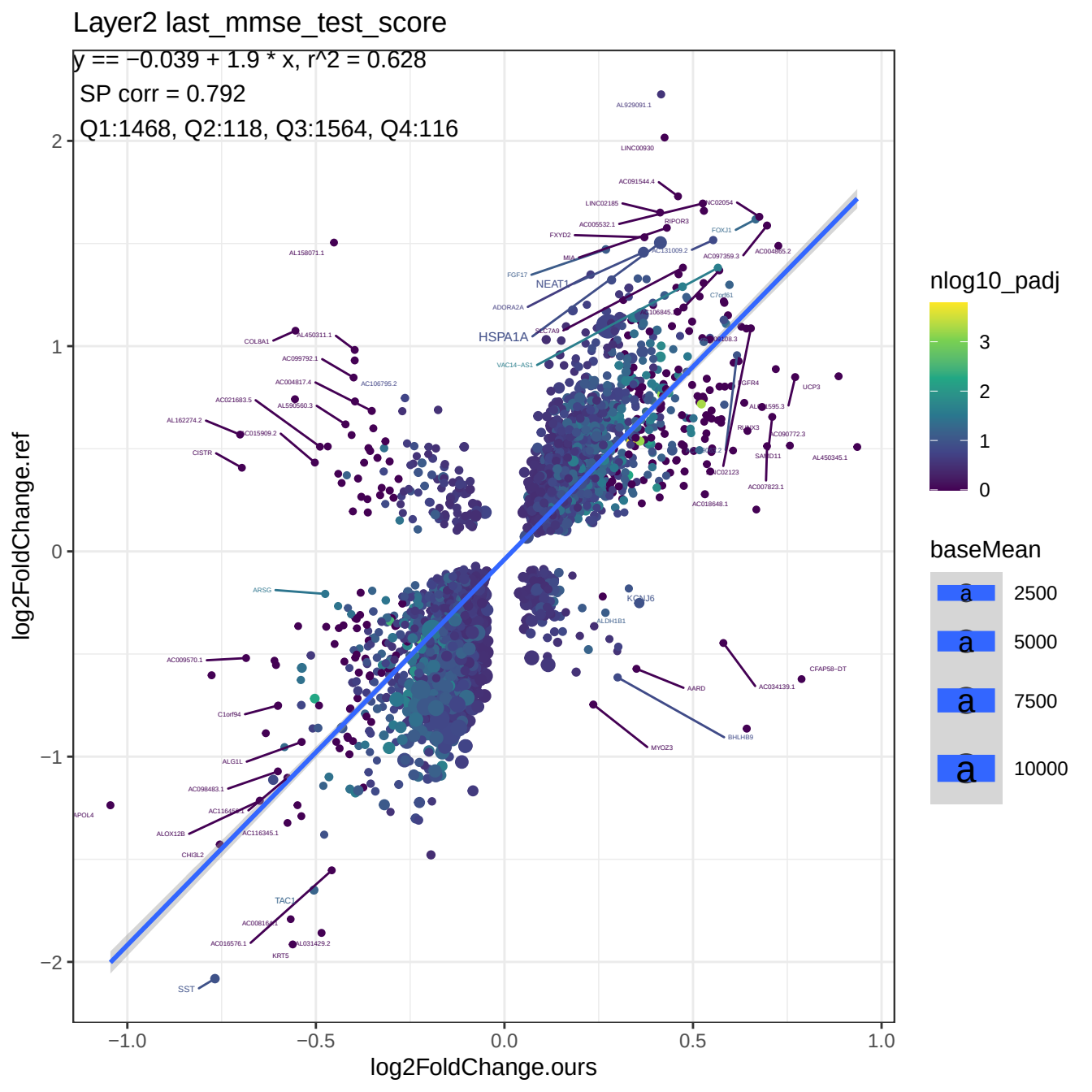
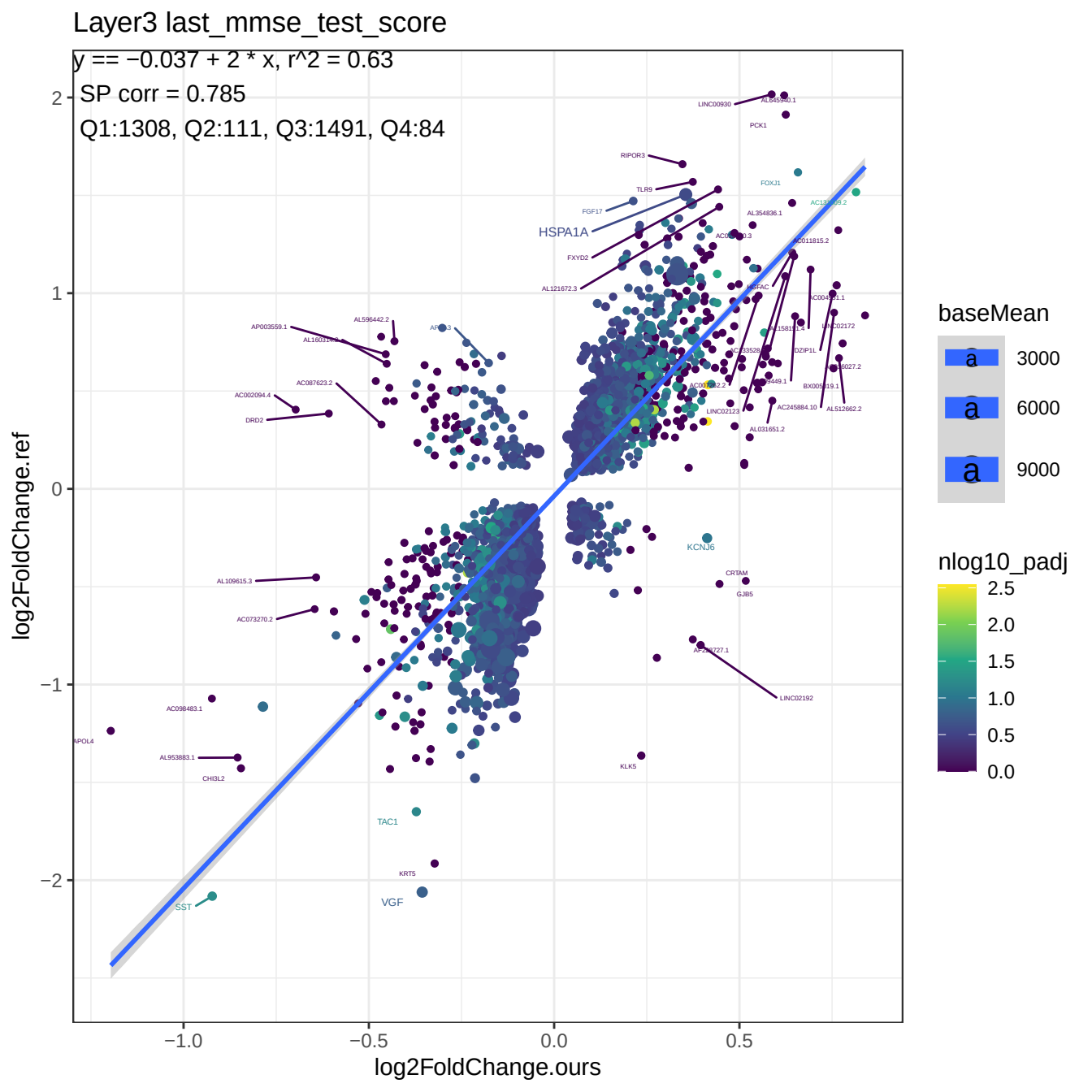
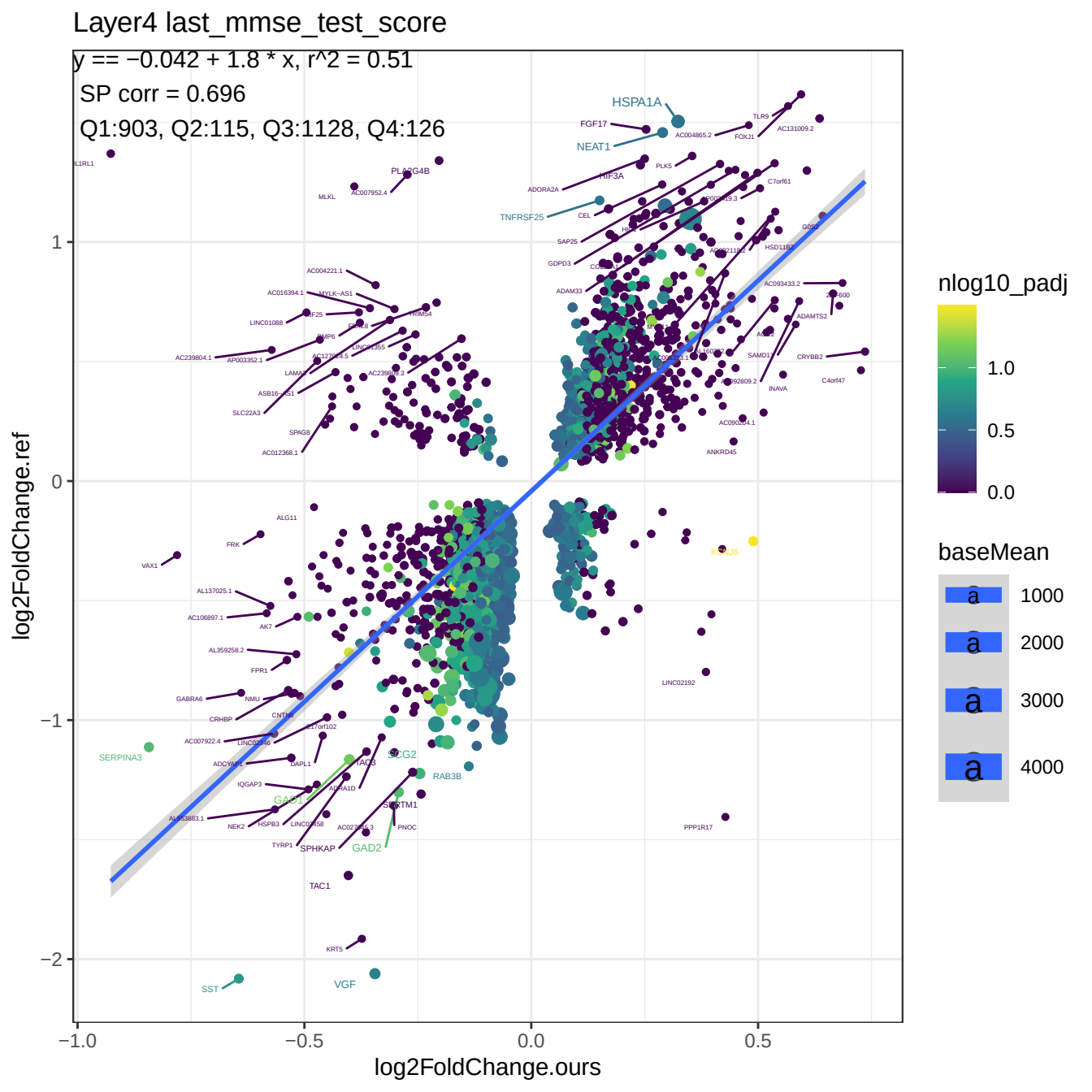
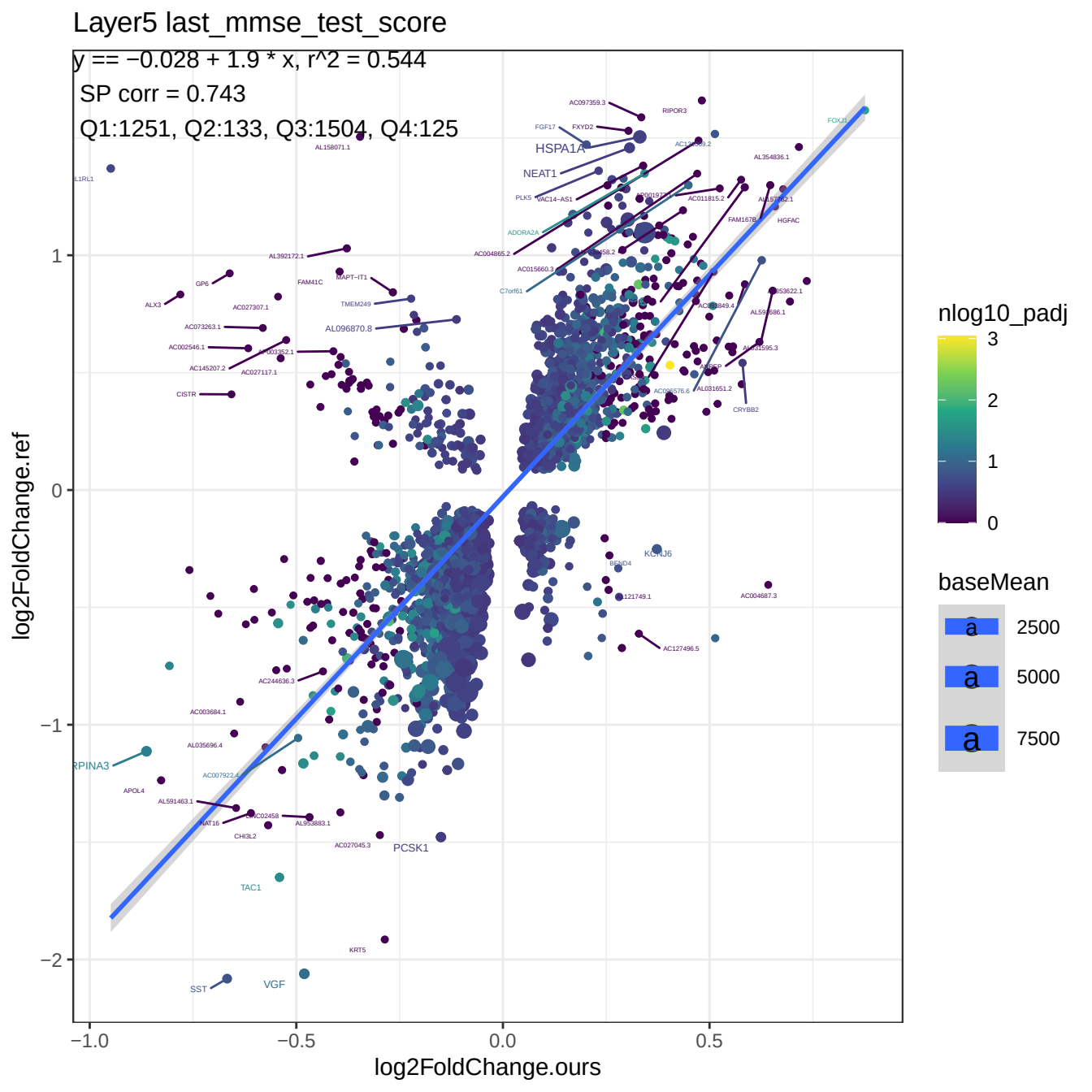
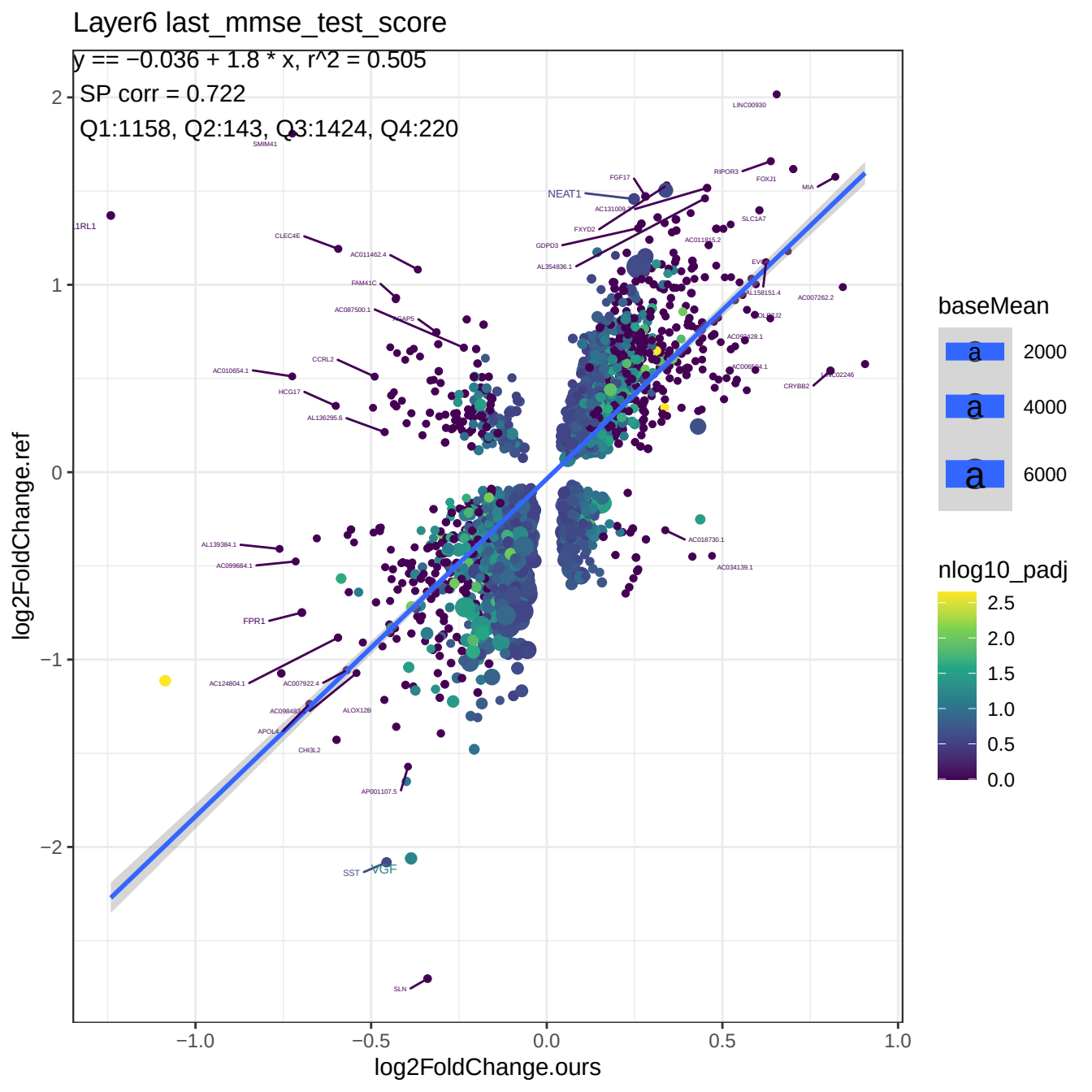
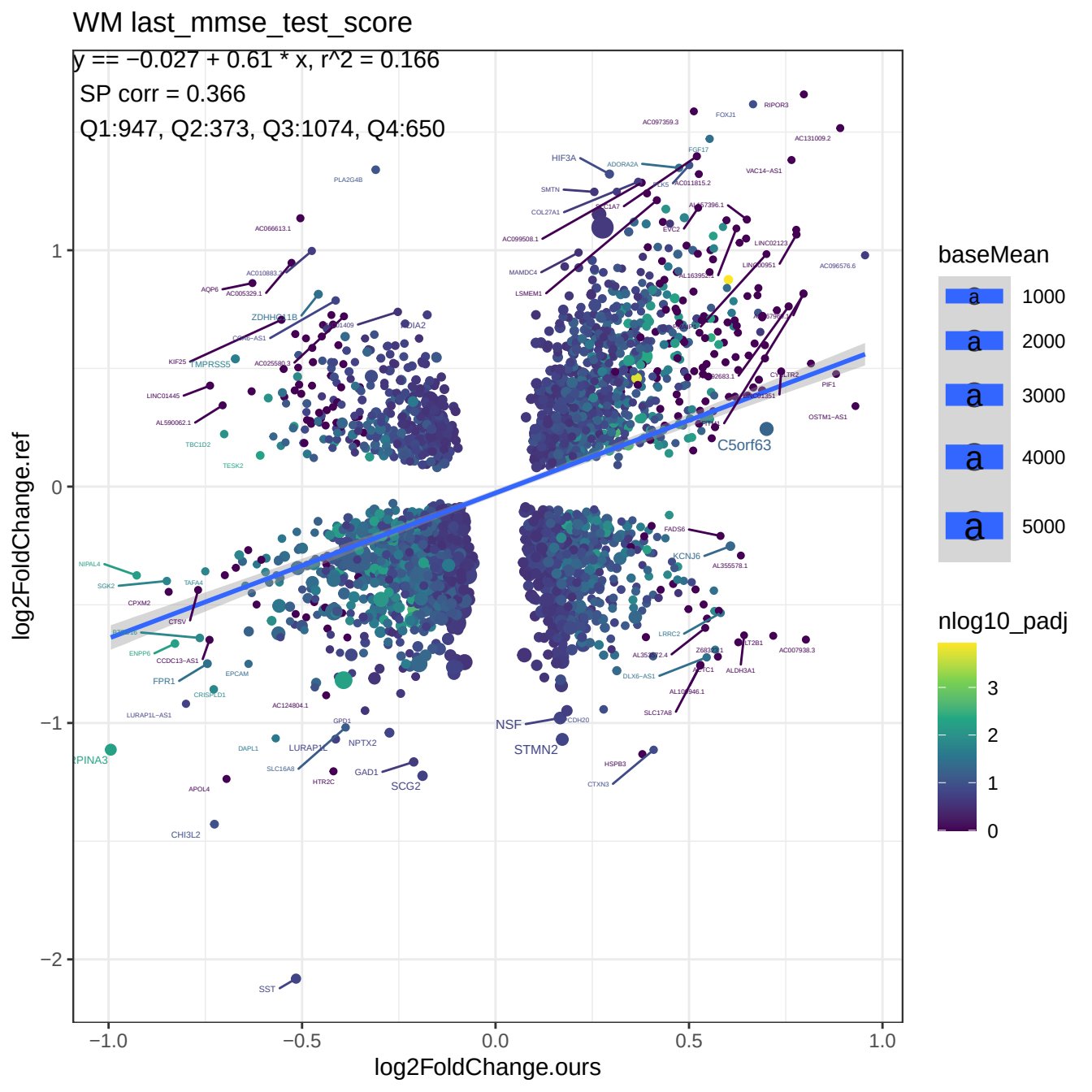

Layer2 last_mmse_test_score
y == −0.039 + 1.9 * x, r^2 = 0.628
SP corr = 0.792
Q1:1468, Q2:118, Q3:1564, Q4:116
log2FoldChange.ref
log2FoldChange.ours
2
1
0
−1
−2
−1.0
−0.5
0.0
0.5
1.0
NEAT1
HSPA1A
TAC1
SST
KRT5
ARSG
KCNJ6
ALDH1B1
AARD
MYOZ3
BHLHB9
CFAP58−DT
AC034139.1
ALOX12B
CHI3L2
APOL4
ALG1L
C1orf94
AC009570.1
AC098483.1
AC116345.1
AC016576.1
AL031429.2
AC008163.1
CISTR
AC015909.2
AL162274.2
AC021683.5
AL590560.3
AC004817.4
AC099792.1
AL450311.1
COL8A1
AC106795.2
AL158071.1
FGF17
ADORA2A
VAC14−AS1
SLC7A5
AC131009.2
AC097359.3
AC004865.2
FOXJ1
FXYD2
AC005532.1
RIPOR3
LINC02185
AC091544.4
LINC00930
AL929091.1
AC018648.1
AC007823.1
AL450345.1
SAMD11
C090772.3
RANX3
AL031595.3
FGFR4
UCP3
nlog10_padj
3
2
1
0
baseMean
2500
5000
7500
10000


Layer3 last_mmse_test_score
y == −0.037 + 2 * x, r^2 = 0.63
SP corr = 0.785
Q1:1308, Q2:111, Q3:1491, Q4:84
log2FoldChange.ref
log2FoldChange.ours
2
1
0
−1
−2
−1.0
−0.5
0.0
0.5
baseMean
3000
6000
9000
nlog10_padj
2.5
2.0
1.5
1.0
0.5
0.0
HSPA1A
VGF
SST
TAC1
KRT5
KCNJ6
CRTAM
GJB5
KLK5
LINC02192
LINC00930
PCK1
RIPOR3
TLR9
FGF17
FXYD2
FOXJ1
AL354836.1
AL121672.3
AL596442.2
AP003559.1
AC087623.2
AC002094.4
DRD2
AL109615.3
AC073270.2
AC098483.1
AL953883.1
CHI3L2
APOL4
LINC02123
AL031651.2
AC245884.10
AL512662.2
BX005219.1
DZIP1L
LINC02172
AC011815.2


Layer4 last_mmse_test_score
y == −0.042 + 1.8 * x, r^2 = 0.51
SP corr = 0.696
Q1:903, Q2:115, Q3:1128, Q4:126
log2FoldChange.ref
log2FoldChange.ours
1
0
−1
−2
−1.0
−0.5
0.0
0.5
HSPA1A
FGF17
NEAT1
TLR9
AC131009.2
AC004865.2
FOXJ1
PLK5
HIF3A
ADORA2A
C7orf61
TNFRSF25
CEL
SAP25
GDPD3
HSD11B2
AC093433.2
ADAM33
ADAMTS2
CRYBB2
C4orf47
INAVA
SAMD12
ANKRD45
AC090204.1
AC092809.2
KCNJ6
LINC02192
PPP1R17
PLA2G4B
MLKL
AC007952.4
AC004221.1
AC016394.1
MYLK-AS1
LINC01088
TRIM54
AC239804.1
AP003352.1
LAMA5
AC239868.3
ASB16-AS1
SLC22A3
SPAG8
AC012368.1
ALG11
FRK
VAX1
AL137025.1
AC106897.1
AK7
AL359258.2
FPR1
GABRA6
NMU
CRHBP
CNTN4
C17orf102
AC007922.4
LINC02346
SERPINA3
ADCYAP1
DAPL1
IQGAP3
TAC3
SCG2
RAB3B
ADRA1D
SORTM1
GAD1
AL033383.1
NEK2
HSPB3
LINC00458
AC027763.3
PNOC
TYRP1
SPHKAP
GAD2
TAC1
KRT5
SST
VGF
nlog10_padj
1.0
0.5
0.0
baseMean
a
1000
2000
3000
4000


Layer5 last_mmse_test_score
y == −0.028 + 1.9 * x, r^2 = 0.544
SP corr = 0.743
Q1:1251, Q2:133, Q3:1504, Q4:125
log2FoldChange.ref
log2FoldChange.ours
1
0
−1
−2
−1.0
−0.5
0.0
0.5
HSPA1A
NEAT1
PCSK1
TAC1
VGF
SST
KRT5
RPINA3
FOXJ1
HGFAC
CRYBB2
KCNJ6
nlog10_padj
3
2
1
0
baseMean
a
2500
5000
7500


Layer6 last_mmse_test_score
y == −0.036 + 1.8 * x, r^2 = 0.505
SP corr = 0.722
Q1:1158, Q2:143, Q3:1424, Q4:220
log2FoldChange.ref
log2FoldChange.ours
2
1
0
−1
−2
−1.0
−0.5
0.0
0.5
1.0
baseMean
a
2000
4000
6000
nlog10_padj
2.5
2.0
1.5
1.0
0.5
0.0
LINC00930
SMIM41
FGF17
RIPOR3
FOXJ1
MIA
NEAT1
AC131009.1
SLC1A7
FXYD2
GDPD3
AC011445.2
AL354836.1
EVC
AL158151.4
AC007262.2
CLEC4E
AC011462.4
FAM41C
AC087500.1
CAP5
AC092428.1
CCRL2
AC010654.1
HCG17
AL136295.6
CRYBB2
AC002246
AL139384.1
AC099684.1
AC018730.1
AC034139.1
FPR1
AC124804.1
AC007922.4
ALOX12B
APOL4
CHI3L2
AP001107.5
SST
VGF
SLN


WM last_mmse_test_score
y == −0.027 + 0.61 * x, r^2 = 0.166
SP corr = 0.366
Q1:947, Q2:373, Q3:1074, Q4:650
log2FoldChange.ref
log2FoldChange.ours
1
0
−1
−2
−1.0
−0.5
0.0
0.5
1.0
C5orf63
NSF
STMN2
SST
SCG2
GAD1
NPTX2
LURAP1L
HIF3A
ADORA2A
PLA2G4B
TMPRSS5
KIF25
ZDHHC11B
DIA2
AQP6
TESK2
TBC1D2
LINC01445
AL590062.1
SGK2
NIPAL4
TAFA4
CPXM2
CTSV
ENPP6
CCDC13−AS1
FPR1
EPCAM
CRISPLD1
LURAP1L−AS1
AC124804.1
GPD1
DAPL1
SLC16A8
HTR2C
APOL4
CHI3L2
KCNJ6
FADS6
AL355578.1
LRRC2
DLX6−AS1
SLC17A8
HSPB3
CTXN3
ALDH3A1
AC007938.3
baseMean
1000
2000
3000
4000
5000
nlog10_padj
3
2
1
0